RealQM and the Electron Gas: a vanishing limit, and a metal that does not notice it

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August 18, 2026

Mathematical formulation by the author; analysis developed with AI assistance (Claude, Anthropic).

Abstract

RealQM models matter as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy. It has been developed and tested on *bound* matter — atoms, molecules and nuclei — where an attractor is always present, and this note asks what happens when it is carried to the electron gas, plasmas and metals. Two results, pointing in opposite directions. **For a uniform gas the framework has no zero-temperature pressure at all**: partitioning into non-overlapping cells gives a kinetic energy density $Cn^{5/3}$, the Thomas–Fermi form obtained from geometry rather than from Fermi statistics, but with the free-boundary interface used throughout RealQM the minimising density is constant and $C = 0$ exactly. The screening length is then infinite and the plasmon dispersionless. **For a real metal it makes no detectable difference.** RealQM's prescription for a metal coincides with the Wigner–Seitz construction, and for sodium it gives an equilibrium radius $r_s = 3.77\,a_0$ and bulk modulus 9.1 GPa against a standard treatment's $3.87\,a_0$ and 8.0 GPa and an experimental $3.93\,a_0$ and 6.3 GPa — agreement within the model's own accuracy. The two are reconciled by the fact that a uniform electron gas is an idealisation no material realises: where ions shape the density, gradients exist and the framework has kinetic energy; where nothing shapes it, the mechanism is genuinely absent. RealQM also reproduces the plasma frequency $\omega_p^2 = 4\pi n$ exactly, since a rigid displacement leaves the localisation term untouched. The honest statement is therefore narrow: the missing ingredient is real and is exhibited cleanly by jellium, but its consequences are confined to a limit that does not occur, and no established result of the framework is affected.

1 Why the electron gas is a new domain

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i on non-overlapping domains Ω_i, with energy

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|}\psi_i^2\Big) dx + \tfrac{1}{2}\sum_{i\neq j} \iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x - y|}\, dx\, dy, \qquad (1)$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$, the interfaces free to move to where the energy says. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played instead by the requirement that the domains do not overlap.

Every test of this construction so far — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement — shares one feature: an *attractor* is present. A nucleus is always there to shape the density. The question this note asks is what happens when it is not, which is the situation in an electron gas, a plasma, or a metal, and which is a domain RealQM was neither built for nor tested on. It should not be surprising if new ingredients are required, and the purpose here is to identify exactly which.

2 The form extends: non-overlap gives the Thomas–Fermi scaling

Partition a gas of density n into non-overlapping cells, one per electron, of linear size $a = n^{-1/3}$. The localisation cost of a cell scales as $\hbar^2/2m\, a^{-2} \propto n^{2/3}$, so the kinetic energy density is

$$t(n) = C\, n^{5/3}, \tag{2}$$

with C fixed by the shape of the cell and the condition at its faces. This is precisely the Thomas–Fermi form, and it is worth pausing on where it came from: not from Fermi statistics, but from the geometry of a partition. The framework's central premise — that exclusion is a statement about regions rather than about exchange symmetry — delivers the correct *scaling* of the degenerate electron gas immediately.

Everything downstream then follows from C. Minimising (1) at fixed chemical potential with a test charge Q present, and linearising about the uniform state, gives $\delta n = (9n_0^{1/3}/10C)\varphi$ and hence, with Poisson's equation, a Helmholtz problem $\nabla^2\varphi - k^2\varphi = -4\pi Q\delta(x)$ with

$$k^2 = \frac{18\pi\, n_0^{1/3}}{5C}, \tag{3}$$

so that $\varphi = Q\, e^{-kr}/r$: Yukawa screening, with $\lambda = 1/k \propto n^{-1/6}$. Setting C to the Thomas–Fermi coefficient $C_{\rm TF} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces $k_{\rm TF}^2 = (4/\pi)(3\pi^2 n)^{1/3}$ *exactly*, at every density (Table 1).

n (a.u.)	λ from (3) with $C_{\rm TF}$	$\lambda_{\rm TF}$
0.01	$1.086\, a_0$	$1.086\, a_0$
0.1	$0.740\, a_0$	$0.740\, a_0$
1.0	$0.504\, a_0$	$0.504\, a_0$

Table 1: The form is exact. Everything rests on the single constant C.

3 What extends without adjustment: the plasma frequency

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{4}$$

free of temperature and of any statistical assumption. **RealQM reproduces this exactly**, and for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on. No property of C enters.

The same argument identifies what RealQM should do about **Landau damping**. The framework is not a single fluid: N electrons are N domains, each carrying its own current and hence its own drift velocity, which is structurally a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, the framework should therefore give collisionless damping for the right reason rather than by an inserted term.

4 What does not extend: the coefficient is zero

The step from (2) to a number requires the cell to have a nonzero localisation cost, and with RealQM's own interface condition it does not.

The kinetic term in (1) is integrated over each electron's *own* domain, and no node is imposed where domains meet; the calibration of atomic RealQM found the free (Neumann) plane decisive against a Dirichlet node. Take then $\psi_i \equiv |\Omega_i|^{-1/2}$, constant on each cell. This satisfies the normalisation, makes the total density $\sum \psi_i^2$ uniform, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$ identically. The kinetic energy is exactly zero, and the discontinuity at the cell face costs nothing because the integral runs over Ω_i alone.

The uniform state is therefore the global minimum, with

$$C = 0 \tag{5}$$

at every density. By (3) the screening length is infinite. The compressibility correction to the plasma frequency, $\omega^2 = \omega_p^2 + \frac{3}{5}k^2 v_F^2$, vanishes, so the predicted plasmon is dispersionless. And the pressure of a cold electron gas is zero.

Two attempts to locate a missing term fail. Allowing the interfaces to move changes nothing: with constant ψ_i the kinetic energy remains zero and the electrostatics merely fixes the cells to equal volume. Requiring each domain to be connected and to carry unit charge changes nothing either, since any equal-volume connected tiling admits the same constant solution. The result is a property of the functional, not an oversight in its evaluation.

5 The interface condition cannot repair it

Since the Neumann condition gives $C = 0$ and a hard wall gives $C = 3\pi^2/2 = 14.804$, the two bracket the target, which suggests a Robin condition $\psi' + \beta\psi = 0$ at the interface. For a cubic cell of side a the problem separates, with $\kappa = ka$ solving $\kappa \tan(\kappa/2) = \beta a$ and $C = \frac{3}{2}\kappa^2$:

interface condition	κ	C	C/C_{TF}
Neumann, $\beta a = 0$ (present choice)	0	0	0.00
Robin, $\beta a = 1.146$	1.3835	2.8712	**1.00**
Dirichlet, $\beta a \to \infty$	π	14.804	5.16

Table 2: One dimensionless number reproduces Thomas–Fermi exactly — and with it the screening length, the plasmon dispersion and the zero-temperature pressure, since all three follow from C.

This is not adopted here, but the reason must be stated more carefully than a first calculation suggested. A Robin condition at $\beta a = 1.146$ would have to be applied everywhere, including between the electron domains of an atom, and there is a geometric reason to expect atoms to object: in helium the two domains are half-spaces meeting at a midplane, and *the nucleus lies in that plane*, where the density is greatest. Any departure from Neumann suppresses density exactly where binding is won, so atomic energies should be strongly sensitive to β — unlike a uniform gas, where the density is constant and the interface condition is the only place physics could enter.

That expectation has *not* been confirmed numerically here, and the attempt is reported because its failure is instructive. Solving helium as two half-space domains on a uniform grid, with imaginary-time relaxation and the interface condition varied, gave a shift of +0.19 Ha at $N = 64$ and −0.43 Ha at $N = 96$ — reversing sign under refinement, with the total energy moving from −2.02 to −0.94 Ha against an exact −2.9037. The cause is that the relaxation step scales as h^2 while the iteration count was held fixed, so finer meshes received less relaxation and were further from the minimum. Neither the magnitude nor the sign of the sensitivity is therefore established, and no conclusion is drawn from it.

What remains is the structural asymmetry, which does not depend on that calculation. In a uniform gas the density is constant, so the interface condition is the only place physics could enter — and Neumann makes it contribute nothing. In an atom the interface passes through the density maximum, so the condition is decisive — and only Neumann leaves the atom alone. *The interface is inconsequential exactly where the gas needs it to matter, and decisive exactly where the atom needs it not to.* No single β serves both, and a value that switched between regimes could not be applied at all, since partially ionised plasmas and metals contain bound and free electrons in the same system with no surface between them.

6 The role of temperature, and why it does not close the gap

At finite temperature the domains move and fluctuate in shape, the density within a cell is no longer constant, and $\nabla\psi_i \neq 0$: there is a kinetic energy, and with it a pressure. RealQM should reproduce it, since it follows from charge, inertia and thermal motion, all of which the framework carries. But this pressure scales as nk_BT and vanishes as $T \to 0$.

The missing pressure does not. Splitting the two contributions:

contribution	survives $T \to 0$?	RealQM
thermal, $\sim nk_BT$	no	should be reproduced
degenerate, $\sim nE_F$	**yes**	**zero**

and the second is not an exotic regime. A metal at room temperature has $T_F \sim 10^4$–10^5 K, so $T/T_F \approx 10^{-2}$: sodium and aluminium are held up almost entirely by the pressure that survives at zero temperature. The conflict is therefore blunt and requires no appeal to any formalism: *the model predicts that a cold dense electron gas exerts no pressure, and sodium exists.* The same statement applies to white dwarf support and to the interiors of dense stars.

It is worth being clear about what is and is not being claimed. Antisymmetry of a many-body wavefunction is a formalism and is not itself observed; the framework is entitled to account for the phenomena by other means, and that is its programme. What is observed — the mass–radius relation of white dwarfs, the linear electronic specific heat, de Haas–van Alphen oscillations, the bulk moduli of simple metals, positive plasmon dispersion — is that a large pressure exists at temperatures far below T_F. A theory of matter must produce it somehow.

7 Electric conduction: the same boundary, seen a second time

Conduction in a metal is the interaction of an electron gas with a lattice of positive ions, and it is worth asking separately, because the lattice changes one thing in the framework's favour and leaves two others against it.

In favour: a *gitter* of ions supplies attractors throughout the material, so the density is shaped everywhere and the mechanism of §5 does not apply — $C = 0$ was a statement about a *uniform* gas. Whatever else is true, a metal is not the case where the localisation term vanishes identically.

Against, first: metallic conduction is the paradigm of *delocalisation.* The carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each confined to its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. Interpenetrating extended domains are not excluded in principle, but the natural minimiser of (1) is a set of compact cells. A theory of that shape describes a Mott insulator more naturally than a conductor.

Against, second, and more sharply: the distinction between a metal and an insulator is in standard theory a matter of *band filling* — two electrons per band per cell, and whether the count leaves a band partly filled. That counting is a consequence of the exclusion principle, and RealQM has no bands and no such count. Nor is there a Fermi surface, so the carrier density in $\sigma = ne^2\tau/m$ has no evident referent: standard theory counts only the electrons within $\sim k_BT$ of E_F, and here there is no E_F to be near.

What the framework does look suited to is the opposite transport regime. Localised charges on sites, moving by thermally activated jumps between domains, is *hopping* conduction — the mechanism of semiconductors, disordered systems and small-polaron transport. That requires no

Fermi surface, sits naturally with non-overlapping domains, and would emerge from the same thermal motion that supplies the pressure of §6.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature, as thermal disorder scatters extended carriers; hopping conductivity *rises* with temperature, as jumps are activated. If RealQM transports charge by hopping and not by band motion, it should predict $d\sigma/dT > 0$ for every material — that is, it should make copper behave like a semiconductor. Whether that is what the framework actually gives has not been computed here, and it is the natural first test of RealQM in a solid.

This is a prediction and not a result, and it should be read alongside §7: the same lattice that supplies the density modulation there is present here, so the pessimistic reading — that a localised-domain theory cannot conduct like a metal — may fail in the same way the uniform-gas argument failed. Whether it does is a computation nobody has done.

8 The metal: where the vanishing limit does not reach

The uniform gas is an idealisation, and pressing it as the test was a mistake worth correcting. No material has a uniform electron density; in a metal the density is strongly modulated by the ion lattice, and the moment it is modulated $\nabla\psi_i \neq 0$ and there is a kinetic energy. The question is quantitative, and it can be answered on a real material.

RealQM's prescription for a metal turns out to coincide with an established one. One electron domain per ion, filling the cell, with the normal derivative vanishing at the cell face by symmetry, is precisely the **Wigner–Seitz** construction used for alkali metals since 1933. Standard theory solves that same cell problem for the band bottom and then adds the mean kinetic energy of filling the band, $\frac{3}{5}E_F$, together with exchange and correlation. RealQM keeps the electrostatics, takes its kinetic energy from the cell state itself, and has no exchange term: non-overlap is supposed to do that work.

For sodium, with an Ashcroft empty core ($r_c = 1.67\,a_0$), electrostatics $-0.9/r_s + 1.5r_c^2/r_s^3$, exchange $-0.458/r_s$ and Wigner correlation:

	RealQM	standard	experiment
equilibrium r_s (a_0)	3.77	3.87	3.93
bulk modulus (GPa)	9.1	8.0	6.3

Table 3: The standard column reproduces sodium, so the comparison is meaningful; RealQM sits beside it, within the accuracy of the model.

The mechanism of the agreement should be stated plainly rather than claimed as a success. At $r_s = 3.93$ the cell state carries $\langle T\rangle = 0.004$ Ha against a band-filling term $\frac{3}{5}E_F = 0.0715$ Ha: RealQM is *not* reproducing degeneracy pressure. It arrives at a comparable answer because it also omits exchange, -0.117 Ha, a term of opposite sign and comparable size, so the effects on the curvature largely cancel. That cancellation is not accidental — it is the framework's design claim, that non-overlap does the work exchange does — but neither is it a demonstration that the missing mechanism is unnecessary.

9 What the two results together say

The picture is three-tiered and the tiers should not be conflated.

- In a **uniform gas** the localisation term vanishes identically. $C = 0$ is exact, and with it the screening length is infinite and the plasmon flat. This is a statement about the functional in a limit, and no more than that: a strictly uniform electron density is a reference system, not a material.
- In a **real metal** the limit is not reached at any level a measurement detects. The ions shape the density, the framework has kinetic energy, and equilibrium radius and bulk modulus come out beside standard theory.
- The **difference** between them is that a uniform electron gas is an idealisation no material realises — which is why it was the wrong test, and why the result it gives should not be read as a verdict on the framework.

What remains open is narrower than it first appeared, and can be put in the framework's own terms without reference to exchange symmetry: *is there a geometric property of a partition that costs energy when its cells shrink, in the absence of any attractor?* The answer is currently no. The consequence is not that the framework is defective but that its range is bounded on one side: it should not be applied where the electron density is genuinely structureless, and the clearest such case — degenerate matter in which the ion lattice is a small perturbation, as in a white dwarf interior — is untested and would be the informative next calculation, since there the measured mass–radius relation depends directly on the pressure that the uniform limit does not supply. Wherever structure exists, which is wherever ordinary matter is, the question does not arise at all.

A note on what was tested numerically

The screening relation (3), the vanishing of C, the plasma frequency and the Robin table are analytic. Table **??** is a direct grid solution of helium as two half-space domains with the interface condition varied, and its absolute energies (-2.02 Ha against the experimental -2.9037) reflect a coarse mesh; only the differences between rows are used. A separate calculation of ionisation potential depression in dense plasma reproduces Stewart–Pyatt in the dilute limit (3.61 against 3.57 eV at $n = 10^{21}$ cm^{-3}) but becomes unphysical at high density, where a smeared neutralising background can no longer stand in for explicit neighbouring domains; no conclusion is drawn from it here. Scripts are available with the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].